

NCAOS-Urielle EGS · Phase 02

Core Shell Engine + API Skeleton · Observer → Judge → Enforcer + Fastify REST/WebSocket

Phase 01 tests	47 / 47	Phase 02 shell	15 / 15	Phase 02 API	8 / 8	Total tests	70 / 70
Manual API tests	5 / 5	Build	npm run build	Git commit	2066733	Files committed	19 files, 2369 ins

@ncaos/api · Fastify REST + WebSocket · v0.2.0

HTTP server :3000 · X-Partner-ID tenant isolation · Watchdog-supervised · 8 tests passing

NEW

Route	Method	Purpose	Key behaviour
/v1/process	POST	Submit boundary signal → Verdict + authority	401 no header · 400 bad Zod · 200 verdict · EVD handle on containment
/v1/state	GET	Current GovState (tenant-scoped)	Full GovState object incl. rawExposurePct, operationalMode, integrity
/v1/audit/:handle	GET	Retrieve containment event by EVD handle	404 for unknown · 200 + full GovEvent for known
/v1/health	GET	Watchdog + shell status	status=OK, watchdog=RUNNING, shell=True, policy
/v1/state/stream	WS	Live GovState push to Admin Terminal	WebSocket feed · Phase 05 Admin UI target

@ncaos/shell · Observer → Judge → Enforcer · v0.2.0

Process-isolated shell engine · single ShellLoop.process() entry point · 15 tests passing · ADR-003

NEW

Module	File	Purpose	Notes
Observer	observer/observer.ts	I/O boundary tap. Emits ObservedRequest or null	null → FAIL_SAFE path (ADR-002 invariant)
Judge	judge/judge.ts	Runs all 4 detection layers, aggregates worst severity+scope	Calls @ncaos/core detection functions
Enforcer	enforcer/enforcer.ts	Verdict + GovState builder. Emits shell:output event	WebSocket payload origin
ShellLoop	shell-loop.ts	Wires Observer→Judge→Enforcer. Single process() entry	15 integration tests
Watchdog	watchdog/watchdog.ts	3-restart policy, exponential backoff, permanent FAIL_SAFE	ADR-003: process isolation supervisor

Shell test suites: lifecycle (3) · clean request PILOT-OBS (6) · threat request ENTERPRISE_STRICT (5) · event emission (1)

@ncaos/core · Detection + scoring + verdict · v0.1.0

Phase 01 — 47 tests all passing (regression confirmed). Judge delegates all detection calls into this package.

External gate	detectGate()	Flag classification → severity+scope	5 tests ✓
Premise check	detectPremise()	Claim ratio → severity; routes to Arch review	4 tests ✓
Authority check	detectAuthority()	Override attempt → HIGH/sys; legitimate→MED	3 tests ✓
Continuity watch	detectContinuity()	Equivalence test fail rate → severity; always sys	4 tests ✓
Verdict engine	enforce() + failSafeVerdict()	PASS/DOWNGRADE/BLOCK/FAIL_SAFE — deterministic	10 tests ✓
Scoring	computeIntegrity() + computeAuthority()	Score 0–100; policy-threshold driven	16 tests ✓

Manual API tests · live server on DESKTOP-2JNEOET · 5 / 5 PASS

Test	Endpoint	Expected	Actual (key fields)	Result
T-01	GET /v1/health	status=OK, watchdog=RUNNING	status=OK shell=True policy=STRICT-PROD	PASS
T-02	POST /v1/process (clean)	action=PASS or DOWNGRADE	REQ-000002 action=DOWNGRADE authority=ref-only integrity=Available/88 EVD-...	PASS
T-03	POST /v1/process (threat)	action=BLOCK, authority=invalid	REQ-000003 BLOCK invalid integrity=Critical/29 routing=SOC EVD-...	PASS
T-04	GET /v1/audit/:handle	Event with layer, severity, routing	evidenceHandle confirmed event.requestId=REQ-000003 full event returned	PASS
T-05	GET /v1/state	Full GovState post-threat	ENFORCING (STRICT) CONTAINMENT rawExposurePct=100 governedOutputPct=54 SOC routing	PASS

Defects found & fixed during testing — 7 fixes, all verified PASS

ID	File	Root cause	Fix applied	ADR
F-01	shell/vitest.config.ts	Vitest cannot resolve @ncaos/core without dist/	Alias: @ncaos/core → ../core/src/index.ts	—
F-02	api/vitest.config.ts	Same + @ncaos/shell also unresolvable	Aliases for both packages + JS→TS rewrite plugin	—
F-03	watchdog/watchdog.ts	Import ./shell-loop.js resolves inside watchdog/	Changed to ../shell-loop.js (one level up)	ADR-003
F-04	api/tsconfig.json	Missing tsconfig — tsc printed help, exit 1	Created tsconfig.json extending tsconfig.base.json	—
F-05	api/src/server.ts:71	TS7016: no declaration file for module ws	npm install --save-dev @types/ws	—

F-06	api/src/server.ts:167	TS2379: exactOptionalPropertyTypes on 4 Zod fields	Changed .optional() → .default() on workflowId, updateId, equivalenceTestResults, continuityScore	ADR-001
F-07	api/package.json	MODULE_NOT_FOUND: dist/index.js does not exist	start script: dist/index.js → dist/server.js	—

Full automated test results · 70 / 70 passing

Package	File	Suite	Test s	Status
@ncaos/core	scoring.test.ts	computeIntegrity	5	PASS
		computeAuthority	4	PASS
		computeContainmentMetrics	3	PASS
		scoreToContinuityState	4	PASS
	detection.test.ts	detectGate	5	PASS
		detectPremise	4	PASS
		detectAuthority	3	PASS
		detectContinuity	4	PASS
	verdict.test.ts	enforce (all profiles) + failSafe + policy	15	PASS
@ncaos/shell	shell-loop.test.ts	ShellLoop lifecycle	3	PASS
		clean request — PILOT	6	PASS
		threat request — ENTERPRISE_STRICT	5	PASS
		event emission	1	PASS
@ncaos/api	api.test.ts	GET /v1/health	1	PASS
		POST /v1/process	4	PASS
		GET /v1/state	1	PASS
		GET /v1/audit/:handle	2	PASS
TOTAL			70	ALL PASS ✓

New architecture decision record

ADR-003	Process isolation invariant	EGS shell runs as a separate process from the AI runtime. A shared-process architecture risks a core crash disabling governance. Watchdog supervisor manages 3-restart policy with exponential backoff. On exhaustion → permanent FAIL_SAFE. Chaos test scheduled Phase 6.
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Phase roadmap

Phase 01	Foundation & repo setup	W1–2	47/47	COMPLETE
Phase 02	Core shell engine + API skeleton	W2–4	70/70	COMPLETE
Phase 03	Detection layer — MTTD instrumentation	W4–6	TBD	NEXT
Phase 04	API & integration surface	W6–8	TBD	PLANNED
Phase 05	Admin UI (production)	W8–10	TBD	PLANNED
Phase 06	Hardening & compliance	W10–12	TBD	PLANNED
Phase 07	Stakeholder demo & handoff	W12–14	TBD	PLANNED